

TurtleBot 4 Resource Guide

Documentation, tools, courses, and help for ROS 2 Jazzy

Student handout | ROS 2 Jazzy on Ubuntu 24.04

This handout collects the resources you will use most with the TurtleBot 4. Each section follows the same pattern: a short explanation, a resource table that says what each resource costs and what to use it for, a table of common issues with fixes, and a practice task. Treat it as a map. The linked pages hold the details, and every link in this PDF is clickable.

Read these five first

#	Resource	Why it comes first
1	Features page of the TurtleBot 4 User Manual turtlebot.github.io/turtlebot4-user-manual/overview/features.html	Know your model (standard or Lite), its sensors, and its speed limits.
2	Basic Setup and Networking turtlebot.github.io/turtlebot4-user-manual/setup/basic.html	How the robot joins Wi-Fi, how to find its IP, and how ROS 2 discovery works.
3	Driving your TurtleBot 4 turtlebot.github.io/turtlebot4-user-manual/tutorials/driving.html	Teleop, velocity commands, and Create 3 actions, with the Jazzy changes.
4	FAQ turtlebot.github.io/turtlebot4-user-manual/troubleshooting/faq.html	Fixes for the problems students hit most often.
5	Generating a map and Navigation turtlebot.github.io/turtlebot4-user-manual/tutorials/generate_map.html	SLAM, saving a map, localization, and Nav2 goals.

JAZZY TABS

Most manual pages have tabs for Galactic, Humble, and Jazzy. This handout uses **Jazzy**. Click the jazzy tab before you copy any command, because launch file names, message types, and firmware versions differ between tabs.

Links were collected in September 2026, from official Jazzy documentation wherever possible. Pages sometimes move. If a link fails, search for the page title on the site shown in the link.

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1 How to use this handout

1.1 Three computers in one robot

A TurtleBot 4 is three computers working together. The **Create 3** base drives the wheels and reads the bumper, cliff, IR, and IMU sensors. The **Raspberry Pi 4B** inside the robot runs the TurtleBot 4 software and the lidar and camera drivers. **Your laptop** runs RViz, SLAM, Nav2, and your own nodes. The Create 3 and the Pi talk over the USB-C cable, which carries both power and an Ethernet link.

In Jazzy the Create 3 stays off Wi-Fi. Its topics live in a hidden namespace called `_do_not_use`, and the `create3_republisher` node on the Pi republishes them under the robot namespace for everyone else. Use the republished names. The `_do_not_use` namespace is only for checking that the base is alive.

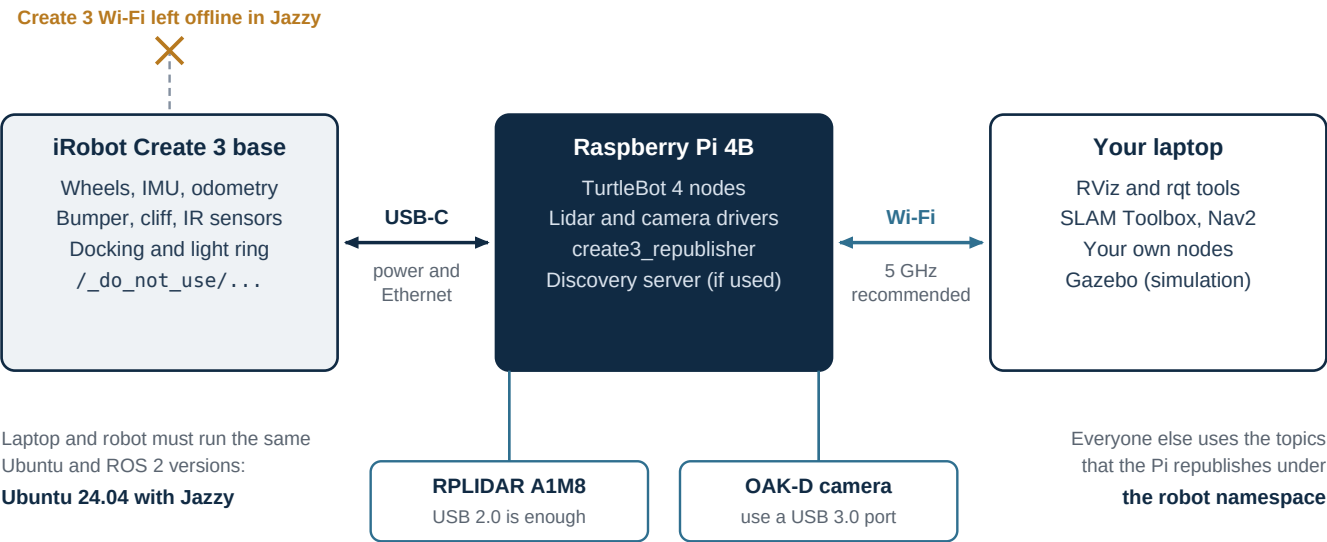


Figure 1. How the three computers connect in the Jazzy setup. Drawn from the Networking and Create 3 pages of the TurtleBot 4 User Manual.

RULE OF THUMB

Most troubleshooting starts with one question: which of the three computers owns the problem? A missing `/odom` points to the Create 3 or the republisher. A missing `/scan` points to the Pi. Topics that exist on the Pi but not on your laptop point to networking.

1.2 Prerequisites

Item	Explain
Ubuntu 24.04	Installed and working on your laptop. The robot runs Ubuntu 24.04, and your laptop must match. releases.ubuntu.com/24.04/
ROS 2 Jazzy	Installed on your laptop, with the talker and listener demo working. docs.ros.org/en/jazzy/Installation/Ubuntu-Install-Debs.html
Laptop Wi-Fi	Must support 5 GHz. The robot access point is a 5 GHz network, and the manual recommends 5 GHz for normal use.
TurtleBot 4	Access to a TurtleBot 4 or TurtleBot 4 Lite running the Jazzy image.

1.3 Your robot's settings

Ask your instructor or lab for these values and write them down. Commands in this handout use `<your_namespace>` and `<your_domain_id>` where they go.

RECORD FOR YOUR ROBOT

Model	Standard or Lite. Decides which manual pages apply and whether you pass <code>model:=lite</code> to launch files.	_____
Namespace	For example <code>/robot1</code> , or none. Put it in front of topic, service, and action names, and pass it to launch files as <code>namespace:=</code> .	_____
Domain ID	Set it with <code>export ROS_DOMAIN_ID=</code> in every terminal on your laptop.	_____
Network setup	Simple Discovery or Discovery Server, and the Wi-Fi network the robot uses.	_____
Robot IP	Shown on the standard model display, or from <code>ros2 topic echo /ip</code> .	_____

2 Official documentation

Two sites cover almost everything: the **TurtleBot 4 User Manual** from Clearpath Robotics and the **Create 3 Docs** from iRobot Education. The manual covers the full robot. The Create 3 site goes deeper on the base: its ROS 2 interface, webserver, firmware, and safety behavior.

Resource	Cost	Use it for
TurtleBot 4 User Manual turtlebot.github.io/turtlebot4-user-manual/	Free	Setup, software, tutorials, and troubleshooting. Start here for any question.
Features and hardware specifications turtlebot.github.io/turtlebot4-user-manual/overview/features.html	Free	Standard vs Lite, sensors, speed limits, battery, and ports.
Manual resources page turtlebot.github.io/turtlebot4-user-manual/overview/resources.html	Free	The manual's own link list for Ubuntu, ROS 2, Create 3, Luxonis, and Slamtec.
TurtleBot 4 Jazzy changelog turtlebot.github.io/turtlebot4-user-manual/changelogs/jazzy.html	Free	What changed for Jazzy: Ubuntu 24.04 image, Create 3 firmware 1.0.0, TwistStamped velocity commands, and gz package names.
Create 3 Docs iroboteducation.github.io/create3_docs/	Free	Base hardware, webserver, ROS 2 APIs, and lessons.
Create 3 ROS 2 interface iroboteducation.github.io/create3_docs/api/ros2/	Free	Every topic, service, and action the base exposes.
Create 3 firmware releases iroboteducation.github.io/create3_docs/releases/overview/	Free	Firmware downloads and release notes.
Clearpath TurtleBot 4 product page clearpathrobotics.com/turtlebot-4/	Free	Product overview from the vendor.

2.1 TurtleBot 4 at a glance

Feature	TurtleBot 4 Lite	TurtleBot 4 (standard)
Size (L x W x H)	342 x 339 x 192 mm	342 x 339 x 351 mm
Weight	3270 g	3945 g
Base and computer	Create 3, Raspberry Pi 4B 4GB	Create 3, Raspberry Pi 4B 4GB
Lidar	RPLIDAR A1M8	RPLIDAR A1M8
Camera	OAK-D-Lite	OAK-D-Pro (adds IR dot projector and IR LED)
Maximum linear velocity	0.31 m/s in safe mode, 0.46 m/s without	Same
Maximum angular velocity	1.90 rad/s	Same
Battery	26 Wh lithium ion, 14.4 V nominal	Same
Operation and charging time	2 h 30 min to 4 h, depending on load; charges in 2 h 30 min	Same
User interface	Create 3 buttons and light ring	Adds a UI board with a 128x64 display, status LEDs, and 4 user buttons
Bluetooth controller	Not included	TurtleBot 4 Controller

Table source: Features page of the TurtleBot 4 User Manual.

Common issue	Fix
Copying a Humble command into a Jazzy system	Click the jazzy tab on every manual page. Jazzy uses <code>turtlebot4_gz_bringup</code> , <code>TwistStamped</code> on <code>/cmd_vel</code> , and <code>view_navigation.launch.py</code> .
Create 3 firmware does not match ROS 2	Jazzy needs firmware whose version starts with I. The Create 3 site lists I.0.0 under Iron, and the TurtleBot 4 manual names it as the firmware for Jazzy.
Base topics vanish after a firmware update	I.* firmware comes in two builds, one for FastDDS and one for CycloneDDS. The Jazzy default is FastDDS. The Pi's <code>RMW_IMPLEMENTATION</code> must match the build on the base.
Following an old blog post or video	Check its ROS version first. Many TurtleBot 4 posts target Humble or Galactic and use old package names.

PRACTICE TASK

Open the Features page and your robot's Create 3 webserver (the Pi IP address with port 8080 in a browser). Write down your robot model, the Create 3 firmware version, and the Pi IP address. Confirm the firmware version starts with I.

3 Setup and networking

Setup has two stages. First the Raspberry Pi joins a Wi-Fi network (Basic Setup). Then the robot and laptops use one of two ways for ROS 2 nodes to find each other (Networking). In most labs the robots are already set up, so you will mostly use these pages to connect your laptop and to recover when something goes wrong.

	Simple Discovery	Discovery Server
How nodes find each other	Multicast. Every participant contacts every other participant.	One or more discovery servers connect publishers and subscribers, similar to the ROS 1 master.
Middleware	FastDDS or CycloneDDS	FastDDS only
Setup	No extra setup; the ROS 2 default	Extra setup on the Pi and on each laptop
Recommended for	1 to 2 robots running simple tasks	Navigation, or 2 or more robots
Watch out for	Multicast problems on university and corporate networks, and high baseline traffic	Every laptop must point to the right server

Table source: Networking page of the TurtleBot 4 User Manual.

bash laptop (from Basic Setup, jazzy tab)

```
# First connection, only for a robot in access point (AP) mode
# Join the 5 GHz Wi-Fi network "Turtlebot4" (password: Turtlebot4)
ssh ubuntu@10.42.0.1          # password: turtlebot4
turtlebot4-setup              # Wi-Fi Setup, set Client mode, save, then Apply Settings
# After the Pi joins your network: find its IP (the standard model also shows it on the display)
source /opt/ros/jazzy/setup.bash
export RMW_IMPLEMENTATION=rmw_fastrtps_cpp
export ROS_DOMAIN_ID=0        # or <your_domain_id> if your lab assigns one
ros2 topic echo /ip
ssh ubuntu@<robot_ip>
```

SHARED ROBOTS

Do not change Wi-Fi, namespace, domain ID, or discovery settings in turtlebot4-setup unless your instructor or lab manager asks you to. Robots are often shared between teams, and one changed setting can take the robot offline for everyone.

Resource	Cost	Use it for
Basic Setup turtlebot.github.io/turtlebot4-user-manual/setup/basic.html	Free	AP mode, Wi-Fi setup, finding the IP, the Create 3 webserver, firmware updates, and recovery over Ethernet.
Networking turtlebot.github.io/turtlebot4-user-manual/setup/networking.html	Free	Choosing Simple Discovery or Discovery Server.
Simple Discovery turtlebot.github.io/turtlebot4-user-manual/setup/simple_discovery.html	Free	Configuring the robot and laptop for multicast discovery.
Discovery Server turtlebot.github.io/turtlebot4-user-manual/setup/discovery_server.html	Free	Configuring the Pi server and each laptop as a client.

Resource	Cost	Use it for
TurtleBot 4 Setup tool turtlebot.github.io/turtlebot4-user-manual/software/turtlebot4_setup.html	Free	What each turtlebot4-setup menu does, plus the image install scripts.
Fast DDS Discovery Server guide fast-dds.docs.eprosima.com/en/latest/fastdds/ros2/discovery_server/ros2_discovery_server.html	Free	How discovery servers work underneath.
ROS 2 concept: About Domain ID docs.ros.org/en/jazzy/Concepts/Intermediate/About-Domain-ID.html	Free	Why domain IDs keep teams apart. Safe values are 0 to 101.
ROS 2 tutorial: Improved Dynamic Discovery docs.ros.org/en/jazzy/Tutorials/Advanced/Improved-Dynamic-Discovery.html	Free	ROS_AUTOMATIC_DISCOVERY_RANGE and ROS_STATIC_PEERS.
Create 3 webserver iroboteducation.github.io/create3_docs/webserver/overview/	Free	Firmware updates, logs, and restarting the Create 3 application.

Common issue	Fix
The Turtlebot4 access point does not appear	It is a 5 GHz network, so your laptop must support 5 GHz. Wait about 30 seconds after power on. Then follow FAQ item 1: check the Pi power LED, cables near the antenna, and restart.
Wrong Wi-Fi password entered; robot unreachable	Connect with an Ethernet cable. Set your wired IPv4 to manual on the same subnet, then run <code>ssh ubuntu@192.168.185.3</code> and fix the Wi-Fi settings.
<code>ros2: command not found</code>	Run <code>source /opt/ros/jazzy/setup.bash</code> , then source your workspace <code>install/setup.bash</code> if you built one.
Robot topics are missing on your laptop	Match ROS_DOMAIN_ID, RMW_IMPLEMENTATION, and discovery settings with the robot. Campus networks often block multicast, so check which network setup your robot uses (section 1.3).
Create 3 topics such as <code>/odom</code> are missing everywhere	On the Pi, run <code>ros2 topic list --include-hidden-topics grep _do_not_use</code> . If the list is empty, tell the instructor. The manual fix is Reset Create3 in turtlebot4-setup, then a reboot.
Another team's robot shows up in your topic list	You share a domain ID. Use the domain ID and namespace assigned to your robot (section 1.3).
An old tutorial says <code>export ROS_LOCALHOST_ONLY=1</code>	That variable was deprecated in Iron. In Jazzy use <code>export ROS_AUTOMATIC_DISCOVERY_RANGE=LOCALHOST</code> to keep traffic on your laptop.

PRACTICE TASK

With your robot's settings, run `ros2 topic list` and `ros2 node list` on your laptop. Find `/scan`, `/odom`, `/battery_state`, and `/ip`, with your namespace in front if your robot uses one. Then draw your own version of Figure 1 with the real IP addresses and network name.

4 Software packages and source code

The TurtleBot 4 software is open source under the Apache 2.0 license and lives in the github.com/turtlebot organization. The Pi image already has the robot packages installed. On your laptop, install binary packages with apt. Build from source only when you need to read or change the code, and always use the **jazzy** branch.

bash laptop

```
sudo apt update
sudo apt install ros-jazzy-turtlebot4-desktop          # turtlebot4_viz launch files and RViz configs
sudo apt install ros-jazzy-turtlebot4-navigation       # SLAM, localization, and Nav2 launch files
sudo apt install ros-jazzy-turtlebot4-simulator ros-jazzy-irobot-create-nodes
sudo apt install ros-jazzy-turtlebot4-tutorials       # TurtleBot 4 Navigator examples
sudo apt install ros-jazzy-teleop-twist-keyboard
```

Resource	Cost	Use it for
turtlebot4 (common packages) github.com/turtlebot/turtlebot4	Free	Packages used by both the real and simulated robot, including <code>turtlebot4_description</code> , <code>turtlebot4_msgs</code> , and <code>turtlebot4_navigation</code> .
turtlebot4_robot github.com/turtlebot/turtlebot4_robot	Free	Packages that run only on the robot, such as <code>turtlebot4_bringup</code> .
turtlebot4_desktop github.com/turtlebot/turtlebot4_desktop	Free	Visualization launch files for your laptop (<code>turtlebot4_viz</code>).
turtlebot4_simulator github.com/turtlebot/turtlebot4_simulator	Free	Gazebo Harmonic simulation (<code>turtlebot4_gz_bringup</code>).
turtlebot4_setup github.com/turtlebot/turtlebot4_setup	Free	The setup tool, image scripts such as <code>sd_flash.sh</code> , and udev rules.
turtlebot4_tutorials github.com/turtlebot/turtlebot4_tutorials	Free	Source for the manual tutorials, including the TurtleBot 4 Navigator examples.
turtlebot4_examples github.com/turtlebot/turtlebot4_examples	Free	Extra example code linked from the manual.
turtlebot4-hardware github.com/turtlebot/turtlebot4-hardware	Free	Hardware reference files linked from the manual.
create3_sim github.com/iRobotEducation/create3_sim	Free	Create 3 robot description and simulation packages.
irobot_create_msgs github.com/iRobotEducation/irobot_create_msgs	Free	Message, service, and action definitions for the Create 3.
create3_examples github.com/iRobotEducation/create3_examples	Free	Holds <code>create3_republisher</code> . Its <code>params.yaml</code> lists which base topics the Pi republishes.
TurtleBot 4 Raspberry Pi images download.ros.org/downloads/turtlebot4/	Free	Fresh SD card images for each ROS distro and robot model.
ROS Index index.ros.org	Free	Find which repository and branch a package comes from for Jazzy.

Common issue	Fix
Editing files under /opt/ros/jazzy	Never edit installed packages. Copy the launch or config file into your own package under ~/ros2_ws and pass your copy as a launch argument.
A cloned repository will not build	Clone with -b jazzy. The default branch may target a different ROS distro.
Behavior changed after you built a package from source	A sourced overlay workspace wins over the apt install. Check which copy is active with <code>ros2 pkg prefix <package></code> .
Running <code>apt upgrade</code> or reflashing a shared robot	Ask your instructor or lab manager first. The Pi image is often shared between teams.
Not sure which arguments a launch file takes	Run <code>ros2 launch <package> <file> --show-args</code> before editing anything.

PRACTICE TASK

Run `ros2 pkg prefix turtlebot4_navigation` and `ros2 launch turtlebot4_navigation slam.launch.py --show-args` on your laptop. Then open the jazzy branch of `turtlebot4_navigation/config` on GitHub and find the default `slam.yaml`. Note the map resolution it uses.

5 Driving and the Create 3 interface

All driving goes through the Create 3. You can stream velocity commands, use Create 3 actions for exact moves, or drive with the Bluetooth controller (hold L1 for normal speed or R1 for turbo, then use the left joystick). In Jazzy the velocity topic uses `geometry_msgs/msg/TwistStamped`, which trips up many older tutorials.

bash laptop

```
# Keyboard teleop. Jazzy needs stamped messages.
ros2 run teleop_twist_keyboard teleop_twist_keyboard --ros-args -p stamped:=true
# Exact moves with Create 3 actions (add your namespace in front if you have one)
ros2 action send_goal /undock irobot_create_msgs/action/Undock "{}"
ros2 action send_goal /drive_distance irobot_create_msgs/action/DriveDistance \
  "{distance: 0.5, max_translation_speed: 0.3}"
ros2 action send_goal /rotate_angle irobot_create_msgs/action/RotateAngle "{angle: 1.57}"
ros2 action send_goal /dock irobot_create_msgs/action/Dock "{}"
```

Resource	Cost	Use it for
Driving your TurtleBot 4 turtlebot.github.io/turtlebot4-user-manual/tutorials/driving.html	Free	Keyboard and joystick teleop, /cmd_vel, and Create 3 actions.
Create 3 page in the manual turtlebot.github.io/turtlebot4-user-manual/software/create3.html	Free	Every Create 3 action, message, and service, plus how the republisher works.
irobot_create_msgs github.com/iRobotEducation/irobot_create_msgs	Free	Exact field names for action goals and messages.
Create 3 API: Moving the robot iroboteducation.github.io/create3_docs/api/moving-the-robot/	Free	Velocity commands and their limits.

Resource	Cost	Use it for
Create 3 API: Docking iroboteducation.github.io/create3_docs/api/docking/	Free	Dock and undock actions, dock status.
Create 3 API: Safety iroboteducation.github.io/create3_docs/api/safety/	Free	Safe mode, limits on driving backward, and the safety override parameter.
Create 3 API: Odometry iroboteducation.github.io/create3_docs/api/odometry/	Free	How the base computes odometry.
Creating your first node (Python) turtlebot.github.io/turtlebot4-user-manual/tutorials/first_node_python.html	Free	A first rclpy node for the TurtleBot 4, built step by step. A C++ version is also in the manual.

Common issue	Fix
Teleop runs but the robot does not move	Add <code>--ros-args -p stamped:=true</code> . Jazzy expects TwistStamped on <code>/cmd_vel</code> . Also check the namespace.
<code>ros2 topic pub</code> with Twist is ignored	Publish <code>geometry_msgs/msg/TwistStamped</code> and put the velocity under <code>twist:</code> .
The robot will not go faster than about 0.31 m/s	That is the safe mode limit. Do not change the safety override without instructor approval.
Reversing stops after a short distance	The Create 3 limits driving backward because its cliff sensors are on the front half. Turn around instead of reversing far. See the Safety page.
Wheels stop when the robot is lifted	Kidnap detection disables the motors until the robot is back on the floor. Watch <code>/kidnap_status</code> .
Waiting for an action server to become available...	Wrong action name or missing namespace. List what exists with <code>ros2 action list</code> .
Controller does not connect	Press the home button; the light should turn blue. If not, re-pair it with <code>bluetoothctl</code> as shown in Basic Setup, under Supplemental Configuration.

PRACTICE TASK

Undock the robot, then drive a 1 m square using only the DriveDistance and RotateAngle actions. Record `/odom` with `ros2 bag record` and compare the final odometry pose with where the robot really stopped. Dock the robot when you finish.

6 Sensors

The lidar and camera drivers start on the Pi at boot as part of the TurtleBot 4 service, so you normally subscribe to their topics rather than launch drivers yourself. The Create 3 adds its own safety and motion sensors, which reach the rest of the system through the republisher.

Sensor	Connects to	Where to find the data
RPLIDAR A1M8: 360 degree 2D lidar, 12 m range	Pi, micro USB to USB-A. USB 2.0 is enough.	/scan (sensor_msgs/msg/LaserScan)
OAK-D-Pro (standard) or OAK-D-Lite (Lite): color and stereo depth, on-board AI	Pi, USB-C to USB-A. Use a USB 3.0 port.	Topics under oakd. List them with <code>ros2 topic list grep oakd</code> .
IMU and wheel odometry	Create 3	/imu, /odom
Cliff (4, front half), bumper, wheel drop	Create 3	/hazard_detection
IR proximity on the bumper	Create 3	/ir_intensity
Slip, stall, and kidnap detection	Create 3	/slip_status, /stall_status, /kidnap_status

Create 3 topics reach your laptop only if they are on the republisher list in `create3_republisher/bringup/params.yaml`. Names are shown without a namespace.

Resource	Cost	Use it for
Sensors page turtlebot.github.io/turtlebot4-user-manual/software/sensors.html	Free	Connecting, running, and controlling the lidar and OAK-D, plus the Create 3 sensors.
RPLIDAR A1 product page www.slamtec.com/en/Lidar/A1	Free	Lidar specifications.
rplidar_ros on ROS Index index.ros.org/r/rplidar_ros/	Free	Which driver source each ROS distro uses.
DepthAI ROS docs (v2) docs.luxonis.com/software/ros/depthai-ros	Free	Matches the depthai-ros driver on the robot image.
depthai-ros repository github.com/luxonis/depthai-ros	Free	Driver source, launch files, and examples.
Luxonis documentation docs.luxonis.com	Free	OAK-D-Pro and OAK-D-Lite hardware details and the DepthAI API.
Create 3 hardware overview iroboteducation.github.io/create3_docs/hw/overview/	Free	Where each base sensor sits.
Create 3 API: Hazards iroboteducation.github.io/create3_docs/api/hazards/	Free	How /hazard_detection reports bumps, cliffs, and wheel drops.

Common issue	Fix
OAK-D topics are missing	The camera powers down while the robot is docked. Undock the robot.

Common issue	Fix
/scan has no data or the lidar is not spinning	The driver spins the motor only when /scan has a subscriber. Open RViz or run <code>ros2 topic hz /scan</code> . The /start_motor and /stop_motor services work only with auto standby turned off.
/dev/RPLIDAR does not exist on the Pi	Without the TurtleBot 4 udev rules the lidar shows up as /dev/ttyUSB0. Check the cable, then tell the instructor.
Camera images lag or drop on your laptop	Images need bandwidth. Use a 5 GHz network, view compressed or lower rate topics, or process images on the Pi and send results instead.
A tutorial says to install the DepthAI v3 driver	The robot image uses the v2 driver. Do not install v3 on a shared robot. Use the v2 docs.
The lidar frame looks rotated 90 degrees in RViz	That is expected. The lidar is mounted with a 90 degree yaw and TF accounts for it. Do not "fix" it in your code.

PRACTICE TASK

Undock the robot. Measure the rate of /scan and one OAK-D image topic with `ros2 topic hz`, first in an SSH session on the Pi and then on your laptop. Explain any difference between the two.

7 Visualization and debugging tools

RViz and the ROS 2 command line tools answer most questions about what the robot is doing. Run RViz on your laptop, not on the Pi: the Pi runs Ubuntu Server with no display.

bash laptop

```
ros2 launch turtlebot4_viz view_model.launch.py      # robot model only
ros2 launch turtlebot4_viz view_robot.launch.py     # model plus live sensor data
ros2 launch turtlebot4_viz view_navigation.launch.py # map, costmaps, and Nav2 tools
ros2 launch turtlebot4_viz view_diagnostics.launch.py # rqt_robot_monitor
# Namespaced robot: add namespace:=<your_namespace> to any of the above
ros2 topic hz /scan                                # is it publishing, and how fast?
ros2 run tf2_tools view_frames                       # saves the TF tree as a PDF
ros2 doctor --report                                # network, middleware, and package summary
ros2 bag record /scan /odom /tf /tf_static
```

Resource	Cost	Use it for
Rviz2 page turtlebot.github.io/turtlebot4-user-manual/software/rviz.html	Free	What each turtlebot4_viz launch file shows.
TurtleBot 4 Desktop turtlebot.github.io/turtlebot4-user-manual/software/turtlebot4_desktop.html	Free	Installing turtlebot4_desktop and its launch files.
Diagnostics turtlebot.github.io/turtlebot4-user-manual/troubleshooting/diagnostics.html	Free	rqt_robot_monitor and the turtlebot4_diagnostics node.
ROS 2 Tests turtlebot.github.io/turtlebot4-user-manual/troubleshooting/ros2_tests.html	Free	Built-in tests for checking the robot.

Resource	Cost	Use it for
ROS 2 Jazzy tutorials docs.ros.org/en/jazzy/Tutorials.html	Free	CLI tools, rqt, recording and playing back data with ros2 bag, and tf2.
Common issue	Fix	
RViz says no transform exists between two frames	Set Fixed Frame to a frame that exists (map when mapping, odom otherwise) and check the tree with <code>view_frames</code> .	
RViz is empty for a namespaced robot	Pass <code>namespace:=</code> to the <code>turtlebot4_viz</code> launch file. TF topics are namespaced too.	
RViz will not start over SSH on the Pi	The Pi has no desktop. Run RViz on your laptop.	
The Pi CPU is very busy	Diagnostics can be turned off in <code>turtlebot4-setup</code> (ROS Setup, Bash Setup, DIAGNOSTICS). On a shared robot, ask first.	

PRACTICE TASK

With the robot undocked, run `ros2 run tf2_tools view_frames` and open the PDF. Mark which frames come from the Create 3, which come from the robot description, and which appear only when SLAM or localization runs.

8 Simulation

Simulation lets you test code without a robot or a charged battery. The Jazzy simulator uses **Gazebo Harmonic**, the Gazebo release paired with Jazzy. Jazzy packages and launch files use the `gz` prefix. Names with `ign` or `ignition` belong to Humble.

bash laptop

```
sudo apt install ros-jazzy-turtlebot4-simulator ros-jazzy-irobot-create-nodes
export ROS_AUTOMATIC_DISCOVERY_RANGE=LOCALHOST # keep the simulation on your laptop
ros2 launch turtlebot4_gz_bringup turtlebot4_gz.launch.py # default world: warehouse
ros2 launch turtlebot4_gz_bringup turtlebot4_gz.launch.py slam:=true nav2:=true rviz:=true
ros2 launch turtlebot4_gz_bringup turtlebot4_gz.launch.py model:=lite world:=maze
# Worlds: depot, maze, warehouse. Press Play in Gazebo to start the simulation.
```

Resource	Cost	Use it for
TurtleBot 4 Simulator turtlebot.github.io/turtlebot4-user-manual/software/turtlebot4_simulator.html	Free	Install steps, every launch option, and the list of worlds.
Simulation page turtlebot.github.io/turtlebot4-user-manual/software/simulation.html	Free	The manual's simulation overview.
turtlebot4_simulator repository github.com/turtlebot/turtlebot4_simulator	Free	Source, worlds, and bridge configuration.
Gazebo with ROS: supported pairings gazebo-sim.org/docs/harmonic/ros_installation	Free	Why Jazzy pairs with Harmonic, and how <code>ros_gz</code> is installed.
Gazebo Harmonic tutorials gazebo-sim.org/docs/harmonic/tutorials/	Free	Building worlds and working with the Gazebo GUI.

Resource	Cost	Use it for
Nav2 Getting Started docs.nav2.org/getting_started/	Free	Nav2 on minimal TurtleBot simulations; nav2_bringup includes tb4_simulation_launch.py.
Common issue	Fix	
Gazebo opens but nothing moves	The simulation starts paused. Press Play in Gazebo.	
Instructions use turtlebot4_ignition_bringup	Those are Humble instructions. Jazzy uses turtlebot4_gz_bringup and turtlebot4_gz.launch.py.	
The simulated robot and a real robot both show up	Never mix simulated and real robots in one ROS system. Keep the simulation local with ROS_AUTOMATIC_DISCOVERY_RANGE=LOCALHOST or a separate domain ID.	
Your own world is not found	Build turtlebot4_simulator from source, add the world next to the others, and pass world:= and map:=.	
Your node's timestamps do not match the simulation	Set use_sim_time to true for your own nodes.	

PRACTICE TASK

Launch the maze world with SLAM and RViz, drive it with teleop, and save the map. Then relaunch with localization:=true nav2:=true and map:= pointing to your map, set the initial pose, and send one Nav2 goal.

9 Mapping and navigation

The turtlebot4_navigation package wraps SLAM Toolbox and Nav2 in launch files set up for the robot. Use SLAM to build a map, then localization on that map for repeatable navigation. The manual recommends running synchronous SLAM on your laptop rather than the Pi, for a higher resolution map.

bash laptop

```
# Mapping
ros2 launch turtlebot4_navigation slam.launch.py
ros2 launch turtlebot4_viz view_navigation.launch.py
ros2 service call /slam_toolbox/save_map slam_toolbox/srv/SaveMap "name:
  data: 'my_map'"
# Navigating on a saved map (one command per terminal)
ros2 launch turtlebot4_navigation localization.launch.py map:=/full/path/to/my_map.yaml
ros2 launch turtlebot4_navigation nav2.launch.py
ros2 launch turtlebot4_viz view_navigation.launch.py
# Namespaced robot: add namespace:=<your_namespace> to all three
```

Resource	Cost	Use it for
Generating a map turtlebot.github.io/turtlebot4-user-manual/tutorials/generate_map.html	Free	SLAM launch, driving to build the map, and saving it.
Navigation turtlebot.github.io/turtlebot4-user-manual/tutorials/navigation.html	Free	SLAM vs localization, launching Nav2, and the RViz navigation tools.

Resource	Cost	Use it for
TurtleBot 4 Navigator turtlebot.github.io/turtlebot4-user-manual/tutorials/turtlebot4_navigator.html	Free	Python API built on the Nav2 Simple Commander, with docking and undocking.
Multiple robots turtlebot.github.io/turtlebot4-user-manual/tutorials/multiple_robots.html	Free	Namespaces for several robots on one network.
SLAM Toolbox github.com/SteveMacenski/slam_toolbox	Free	Modes and parameters behind the SLAM launch file.
Default SLAM parameters github.com/turtlebot/turtlebot4/blob/jazzy/turtlebot4_navigation/config/slam.yaml	Free	Starting point for your own copy, passed with <code>params:=</code> .
Nav2 documentation docs.nav2.org	Free	Concepts, configuration, tuning, and plugins.
Nav2 Simple Commander API docs.nav2.org/commander_api/index.html	Free	Methods the TurtleBot 4 Navigator builds on.
Nav2 tutorials docs.nav2.org/tutorials/index.html	Free	Keepout zones, speed limits, collision monitor, docking, and more.

Common issue	Fix
Nav2 ignores goals after localization starts	Set the initial pose first with the 2D Pose Estimate tool in RViz.
The map has smeared or doubled walls	Drive slowly and smoothly, return to the start to close loops, and keep people out of the area while mapping.
<code>save_map</code> fails on a namespaced robot	Call the map saver directly: <code>ros2 run nav2_map_server map_saver_cli -f "map_name" --ros-args -p map_subscribe_transient_local:=true -r __ns:=/namespace</code> .
Localization cannot find the map	Pass the full path to the <code>.yaml</code> file, and keep the <code>.pgm</code> where the <code>.yaml</code> says it is.
Navigation is choppy or stops over Wi-Fi	The manual recommends Discovery Server and a 5 GHz network for navigation.
TurtleBot 4 Navigator examples drive to odd places on the real robot	The example goals are for the simulation worlds. Replace them with points on your own map.

PRACTICE TASK

Map a hallway near the lab, save the map, and localize on it. Write a short TurtleBot 4 Navigator script that undocks, visits three goals, and docks again. Record `/tf`, `/odom`, `/scan`, and `/plan` with `ros2 bag record` for your report.

10 Robot description and TF

Nav2, SLAM, and RViz all depend on the robot description and the TF tree. This section lists where to look when you add a sensor or need to understand a frame.

Resource	Cost	Use it for
turtlebot4_description github.com/turtlebot/turtlebot4	Free	The official TurtleBot 4 xacro files (in the <code>turtlebot4_description</code> folder).
irobot_create_description github.com/iRobotEducation/create3_sim	Free	The Create 3 base model the TurtleBot 4 description includes.
Official tutorial: URDF with robot_state_publisher docs.ros.org/en/jazzy/Tutorials/Intermediate/URDF/Using-URDF-with-Robot-State-Publisher-cpp.html	Free	Canonical walkthrough in C++ and Python.
Articulated Robotics: Coordinate Transforms for Robotics articulatedrobotics.xyz/category/coordinate-transforms-for-robotics	Free	Frames and transforms, the usual URDF blocker.
REP 103 and REP 105 www.ros.org/reps/rep-0103.html	Free	Units, axis conventions, and standard frames. REP 105 is at rep-0105.html .

Common issue	Fix
Two nodes publish the same transform	Keep one source per transform. The Create 3 already publishes the odometry transform, so do not publish your own for the same frames.
A sensor you added has no transform	Add a fixed joint to your description, or run a <code>static_transform_publisher</code> .

11 Courses and tutorials

Paid courses are optional. The free resources cover everything you need to work with the TurtleBot 4. When a course was recorded on an older ROS version, the concepts still carry over; only commands and package names change.

Resource	Cost	Use it for
ROS 2 Jazzy tutorials docs.ros.org/en/jazzy/Tutorials.html	Free	The official path from CLI tools to launch files, tf2, and URDF.
Create 3 lessons: Getting started with ROS 2 iroboteducation.github.io/create3_docs/lessons/ros2/overview/	Free	Short ROS 2 lessons built around the Create 3 base.
TurtleBot4Lessons github.com/turtlebot/TurtleBot4Lessons	Free	Classroom lessons and projects from the TurtleBot team. Marked work in progress and written for older Ubuntu and ROS versions, so adapt commands to Jazzy.
Articulated Robotics articulatedrobotics.xyz	Free	Mobile robot build series: lidar, cameras, ros2_control, SLAM with slam_toolbox, and Nav2.

Resource	Cost	Use it for
Edouard Renard: ROS 2 Nav2 [Navigation 2 Stack], with SLAM and Navigation (Udemy) www.udemy.com/course/ros2-nav2-stack/	About \$10 to \$20 on sale	Nav2 and slam_toolbox step by step in Gazebo, plus the Simple Commander API. Taught with TurtleBot 3 on Humble.
Edouard Renard: Level 2, TF, URDF, RViz, Gazebo (Udemy) www.udemy.com/course/ros2-tf-urdf-rviz-gazebo/	About \$10 to \$20 on sale	xacro, robot_state_publisher, RViz, and Gazebo plugins.

12 Getting help

Work through these steps in order: this handout and the manual FAQ, then a search of existing issues and questions, then your team and your instructor or TA, and only then a new public post.

Resource	Cost	Use it for
TurtleBot 4 FAQ turtlebot.github.io/turtlebot4-user-manual/troubleshooting/faq.html	Free	Access point, Bluetooth, OAK-D, and missing Create 3 topic problems.
Create 3 FAQ iroboteducation.github.io/create3_docs/faq/faq/	Free	Base, firmware, and webserver questions.
turtlebot4 GitHub issues github.com/turtlebot/turtlebot4/issues	Free	Search first. The issue template shows what maintainers need.
create3_docs GitHub issues github.com/iRobotEducation/create3_docs/issues	Free	Known Create 3 base and firmware problems.
Robotics Stack Exchange robotics.stackexchange.com	Free	The Q&A site for ROS questions. ROS Answers moved here in August 2023.
ROS Discourse discourse.openrobotics.org	Free	Announcements, release news, and community discussion.

12.1 How to ask a good question

The TurtleBot 4 issue template asks for the details below. Include them in any question, including questions to your instructor or TA. Most problems are solved just by collecting them.

question template	
Robot model:	TurtleBot 4 Standard or Lite
ROS distro:	Jazzy
Networking:	Simple Discovery or Discovery Server
OS:	Ubuntu 24.04
Installed or built:	installed with apt, or built from source
Package version:	<code>dpkg -s ros-jazzy-turtlebot4-navigation grep Version</code>
Expected behavior:	what should have happened
Actual behavior:	what happened instead
Commands and output:	exact commands and the full terminal output, not a phone photo
Already tried:	what you changed and what each attempt showed

BEFORE YOU POST

Remove Wi-Fi passwords, API keys, and personal information from logs and screenshots.

Appendix A Quick reference

Every command below comes from the jazzy tab of the TurtleBot 4 User Manual or the ROS 2 Jazzy documentation. Add your namespace where your robot uses one.

Task	Command
Set up a terminal	<code>source /opt/ros/jazzy/setup.bash</code> <code>export ROS_DOMAIN_ID=<your_domain_id></code>
Find the robot IP	<code>ros2 topic echo /ip</code>
Log in to the Pi	<code>ssh ubuntu@<robot_ip></code>
Open the Create 3 webserver	<code>http://<robot_ip>:8080</code> in a browser
Check the base is alive (on the Pi)	<code>ros2 topic list --include-hidden-topics grep _do_not_use</code>
Drive with the keyboard	<code>ros2 run teleop_twist_keyboard teleop_twist_keyboard --ros-args -p stamped:=true</code>
Undock and dock	<code>ros2 action send_goal /undock irobot_create_msgs/action/Undock "{}"</code> <code>ros2 action send_goal /dock irobot_create_msgs/action/Dock "{}"</code>
View the robot	<code>ros2 launch turtlebot4_viz view_robot.launch.py</code>
Build a map	<code>ros2 launch turtlebot4_navigation slam.launch.py</code>
Save the map	<code>ros2 service call /slam_toolbox/save_map slam_toolbox/srv/SaveMap "name: {data: 'my_map'}"</code>
Localize and navigate	<code>ros2 launch turtlebot4_navigation localization.launch.py map:=<map.yaml></code> <code>ros2 launch turtlebot4_navigation nav2.launch.py</code>
Visualize navigation	<code>ros2 launch turtlebot4_viz view_navigation.launch.py</code>
Run the simulator	<code>ros2 launch turtlebot4_gz_bringup turtlebot4_gz.launch.py</code>
See the TF tree	<code>ros2 run tf2_tools view_frames</code>
Check diagnostics	<code>ros2 launch turtlebot4_viz view_diagnostics.launch.py</code>
List a launch file's arguments	<code>ros2 launch <package> <file> --show-args</code>

Appendix B Common topics, actions, and services

Names are shown without a namespace. Create 3 names reach your laptop through the republisher.

Name	Type	Comes from
/cmd_vel	geometry_msgs/msg/TwistStamped	You, teleop, or Nav2. Humble used Twist.
/odom	nav_msgs/msg/Odometry	Create 3
/imu	sensor_msgs/msg/Imu	Create 3
/battery_state	sensor_msgs/msg/BatteryState	Create 3
/hazard_detection	irobot_create_msgs/msg/HazardDetectionVector	Create 3
/dock_status	irobot_create_msgs/msg/DockStatus	Create 3
/scan	sensor_msgs/msg/LaserScan	RPLIDAR driver on the Pi
/ip	std_msgs/msg/String	TurtleBot 4 software on the Pi
/tf, /tf_static	tf2_msgs/msg/TFMessage	Create 3, robot_state_publisher, and SLAM or localization
/dock, /undock	irobot_create_msgs/action/Dock irobot_create_msgs/action/Undock	Create 3 actions
/drive_distance, /drive_arc, /rotate_angle	irobot_create_msgs/action/DriveDistance irobot_create_msgs/action/DriveArc irobot_create_msgs/action/RotateAngle	Create 3 actions
/navigate_to_position, /wall_follow	irobot_create_msgs/action/NavigateToPosition irobot_create_msgs/action/WallFollow	Create 3 actions
/navigate_to_pose	nav2_msgs/action/NavigateToPose	Nav2 action
/start_motor, /stop_motor	std_srvs/srv/Empty	RPLIDAR driver services

Sources: Create 3 ROS 2 interface, the Create 3 and Sensors pages of the TurtleBot 4 User Manual, and the TurtleBot 4 Jazzy changelog.